Medhansh Garg

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hackoverflow404.github.io/Portfolio
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Education

University of Illinois, Urbana-Champaign Aug. 2023 – May 2027 (Expected)
- B.S. in Computer Engineering | Relevant Coursework: Data Structures & Algorithms, Computer Systems Engineering, Computer Architecture, Operating Systems, Databases

Skills

- **Languages**: TypeScript, Python, Java, C, C++, SQL, Kotlin, Go
- **Frameworks & Web**: React, Next.js, Node.js, FastAPI, Flask, Spring Boot, Prisma, SQLAlchemy, REST APIs
- **Data & Cloud**: PostgreSQL, SQLite, Supabase, Docker, Kubernetes, AWS, Azure, Git
- **Systems**: RISC-V, QEMU, WebAssembly, Emscripten, Android HALs
- **Infra**: Linux, Docker, Kubernetes, nginx, GitHub Actions CI/CD, Tailscale, MQTT, systemd, Samba

Projects

ECE 391 RISC-V Kernel | *C, RISC-V, QEMU* github.com/HackOverflow404/ece391-kernel-demo
- With a team, wrote a **RISC-V64 kernel** from scratch: **VirtIO** console and block drivers, **KTFS** filesystem, page-table virtual memory, scheduler with kernel threads, user/kernel syscalls, and a Unix-style shell with multi-stage pipelines
- Implemented the **VirtIO** drivers directly against the hardware spec and debugged crashes from raw register state, reasoning about **synchronization** and ordering between timer and device interrupts
- Compiled **TinyEMU** to **WebAssembly** (Emscripten) so the kernel boots live in the browser, with an **xterm.js** shell and filesystem mounted over HTTP: ece-391-kernel-demo.vercel.app

Echo Show 5: LineageOS & Camera Pipeline | *C++, Kotlin, HALs* github.com/HackOverflow404/EchoCameraStreamer
- Ported **LineageOS 18.1** via the **amonet bootrom exploit**; fixed a **MediaTek camera ABI/vtable mismatch** and a defconfig bug
- Built a **Kotlin 720p H.264** webcam app that recovers from a full reboot in **55s**; on-demand pipeline cut idle CPU **12–14%** to **0.1%**
- Added luma-only **denoise + sharpen** (∼1/3 core; optional **RTX 4070** path), and patched Android's **Bluetooth stack** to trim persistent audio backlog, holding speaker latency between **30** and **180ms**

Mainframe: Raspberry Pi Homelab | *Raspberry Pi, Tailscale, Docker, MQTT* Portfolio: Mainframe entry
- Operate a **Pi 4** homelab since May 2025 (17 months, about **97%** uptime) over a 3-device **Tailscale** mesh (no router ports opened): **11** self-hosted services (**Home Assistant** in **Docker**, **MQTT** broker, **Samba** NAS, **systemd** services), about 1TB of a 2TB NAS in use
- Wrote **mainframe-bridge**, a **Python** service publishing Pi health, drives, services and camera to **Home Assistant** via **MQTT** discovery with last-will (**49** entities across 4 devices), so a crash shows unavailable instead of stale
- Built **SECUR-T**, camera fan-out to **go2rtc**/**Home Assistant** and a NAS recorder, **25GB** rolling buffer (about 18h), about **7%** Pi CPU

Experience

Hopcharge, Gurugram, India | *Software Engineering Intern* Jun. 2026 – Aug. 2026
- Built **Grid Ops**, replacing 5 hand-run scripts with a **FastAPI**/**React** (**Vite**/**TS**) service covering demand mapping, churn, forecasting, and pod allocation across **3 Delhi-NCR metros** (about 60 hexes), deployed behind **nginx** on an ARM VM
- Built **VOLT.CV**, a **FastAPI** HR dashboard with database-level **Row-Level Security** tied to Google identities, shipped as a cross-platform desktop build with **PyInstaller** and **GitHub Actions** CI for 4 OS/architectures
- Built the **Ad Engine** (**Next.js**, **Prisma**/**Postgres**) publishing ads to **Meta** and **YouTube**; 2–3 ads/week to **3/day**, about **8**×

CS 124 Honors, Urbana, IL | *Lead Web Developer* Aug. 2025 – Present
- Lead development of the CS 124 platform (**Next.js**, **Supabase**) on **Vercel** with **GitHub Actions** CI; **246-test Vitest**/**Playwright** suite (6 unit, 167 API/DB integration, 73 browser E2E) with E2E gating dev-to-main promotions; serving **300+ students** and **30+ staff**
- Built **CILogon** SSO and a 6-role least-privilege system with temporary role-view access, no permanent escalation; all public pages routed through a server-side API layer so no DB keys reach the client
- Took the course site's **Lighthouse** score from **76** to **99** by cutting bloated assets and render-blocking paths; built a **TOTP** attendance system with one-time codes refreshing every 30 seconds
- Automated student/staff, grading, and event management: batch grading, CSV import, and sprint tracking. Lead an **11-person** dev team through weekly meetings and PR reviews

IoT++, Remote | *Software Engineering Intern* Jun. 2024 – Aug. 2024
- Built a real-time fire/smoke detection pipeline with **YOLOv8**, improving detection accuracy **62%** relative to the base YOLO model and throughput **13**× through **pruning** and **tiling**
- Trained **Random Forests** and **RNNs** to route trucks to weighbridges from live traffic and equipment throughput
- Shipped **NPU**-optimized inference on **12 Orange Pi units** in factory pilot, containerized with **Docker** and **Kubernetes** on **Azure**

Care Health Insurance, Gurugram, India | *Cybersecurity Intern* Jun. 2025 – Aug. 2025
- Ran external recon that uncovered **60+ unauthenticated internet-exposed services**, **30** of which were pulled internal after my findings; bypassed **SSL pinning** and root detection (**Frida**, **JADX**) on the customer mobile app

Leadership & Involvement

Illinois MicroTech, Urbana, IL | *External Vice President* | *EOH Project Lead* Oct. 2025 – Present
- Led a **15-person** team delivering the club's Engineering Open House 2026 showcase to **20,000+ attendees** on a **$250** budget
- Manage the club's marketing chairs, EOH leads, and outreach committee, alongside independent outreach initiatives
- HackerFab: engineering a **3-axis stepper-driven stage** with **sub-nm positioning** for wafer photolithography